

Key Activities in Parallels RAS Management Console

Network & Security

- A **firewall** protects the system (like a security guard for the network).
- The system will be moved to **Azure (cloud)**, but client will use their own firewall.
- Only **3 main doors (ports)** are kept open for users:
 - Port 80 (web access)
 - Port 443 TCP (secure browsing)
 - Port 443 UDP (smooth streaming for apps).
- Traffic first goes to **load balancers** (two servers – one active, one backup). These spread the workload evenly among servers.

3. Gateways & Connection Brokers

- After load balancers, traffic moves to **gateway servers** (2 servers, so no downtime if one fails).
- Then it goes to **connection brokers** – these decide which server the user connects to, and they manage security and resources.
- All parts have **backup/redundancy**, so even if one fails, another one takes over.

4. Session Hosts (Application Servers)

- These are the actual servers where software is installed and used.
- Different servers are dedicated to different apps:
 - “01 & 02” → Drake tax app
 - “03” → LERT app
 - “4A” → QuickBooks + tax software combo
 - 5 large servers → Only QuickBooks (since it needs a lot of memory).

5. QuickBooks Resource Use

- QuickBooks is heavy on memory & processing.
- Each QuickBooks server has **48GB RAM**.
- Total of **250GB RAM** only for QuickBooks servers.

- Shows the company has dedicated a lot of power just for QuickBooks users.

6. User Session Management

- Admins can see who is logged in, on which server, how long, and troubleshoot easily.
- They can even see internet speed/quality for each user.
- If a user cannot connect, the system shows **why** (like wrong password, network issue, or security mismatch).

7. Load Balancing & App Access

- New users are assigned to servers based on **how busy** the server is (CPU, memory, active users).
- Apps are given only to the right users, based on **Active Directory groups** (like office departments).
- Quick setup: Add a user to a group → they instantly get the right apps & permissions.

8. Printing & Scanning

- Users can print or scan from their local device while working on the server.
- Drake app printing was disabled (due to issues).
- Scanning is also supported – users can use their local scanner as if it's connected to the server.

9. Security & Login

- Only people in the company domain can log in.
- Idle sessions close after **5 minutes**.
- Extra security option: **FIPS 140 compliance** (government-level security).
- Login hours can be controlled – for example, only allow certain teams during work hours.

10. Multi-Factor Authentication (MFA)

- System supports multiple MFA apps (Google Authenticator, Microsoft Authenticator, Duo, etc.).
- Currently using **Google Authenticator**.
- Admins can reset MFA without changing the main password.

11. Device Control & Policies

- System can detect if a user logs in from a **company laptop or personal device**.
- Admins can block personal devices if required.
- Very fine-grained controls like:
 - Allow/block USB drives
 - Control multiple monitors
 - Stop users from saving passwords

12. Reports & Monitoring

- Reports show: who logged in, how long, app usage, idle time, etc.
- Helps in **billing contractors** or **auditing**.
- Performance monitor shows CPU use, memory, and helps optimize cost in Azure.

13. Training & Audit

- Parallels offers **free 2-hour certification training** for admins.
- Every admin action is recorded in an **audit trail** for accountability.

14. User Experience

- Admins can check a user's **connection quality, latency, and internet speed**.
- Uses **UDP protocol** for smoother user experience.
- Can detect user's internet issue **before the user complains**.

15. Active Directory (AD)

- AD is like a company directory for users & groups.
- Over **800 groups** exist, for different departments and apps.
- Each folder on file servers has its own security group → ensuring only right people access.

16. Group Policies & Drives

- Group policies = rules for users (e.g., no popup in Adobe Reader, auto logout for LERT users).
- Drives (network folders) are auto-mapped based on group membership.

17. File Servers

- Two main file servers:
 - FS01 → “My CFO” data
 - FS02 → “My Tax Filer” data
- Backup & disaster recovery plan in place.
- Folders organized neatly by year, permissions tightly controlled.

18. Software Setup

- Repository stores all installers (Chrome, Acrobat, Drake, QuickBooks, etc.).
- Drake & Lassert apps are installed on **Windows 11 systems**, pointing to shared folders.
- Special folders (“options” and “Yerum”) are critical for Lassert setup.

19. Lassert Servers

- Each Lassert server requires its **own installation** (~30 mins each).
- But they all connect to the **same shared folder**.
- License is shared across servers → 100 users can connect across all servers.

20. QuickBooks Setup

- Three install options depending on role (server, database, or both).
- Updates should always happen on **file server with DB manager**.
- ⚠️ Never scan the D drive for QuickBooks files → it resets all folder permissions (big security risk).
- Most QuickBooks issues can be fixed by **restarting DB34 service**.
- Windows firewall should be off for QuickBooks to work properly.

21. Conclusion & Next Steps

- Yashodhan appreciated the detailed info.
- Plan: Review recording before next meeting.
- Transition to Azure will use **Azure Site Recovery** (keeps same IP addresses, smooth migration).
- Yashodhan mentioned bad past experience with database transfer → hence cautious with large moves.

Here's the list of **which applications are hosted on which servers**:

Applications & Server Mapping

1. Drake Tax Application

- Hosted on servers: **01 & 02**

2. LERT Software

- Hosted on servers: **03**

3. QuickBooks + Tax Combo (for "My Tax Filer")

- Hosted on servers: **4A** (includes Drake + LERT + some QuickBooks)

4. QuickBooks (for "My CFO")

- Hosted on **5 dedicated servers**
- Each server has **48GB RAM** (total ~250GB memory allocated just for QuickBooks).

5. Lassert (Tax Software)

- Runs on **multiple servers**
- Each server has its own installation (~30 mins setup each).
- All point to the same **shared folder** for data.
- **100-user license shared across all Lassert servers** (not per server).

6. **Drake + Lassert (for Windows 11 workstations)**

- Installed on **Windows 11 systems**
- Files stored in **shared tax folders** (2003–2024).

7. **File Servers**

- **FS01** → For “My CFO” data
- **FS02** → For “My Tax Filer” data

8. **Software Repository (Installers)**

- Contains **Acrobat Reader, Chrome, CrowdStrike, Drake, LERT, QuickBooks** (various versions).
- Central place to deploy apps to servers.

In summary:

- **Drake → Servers 01, 02**
- **LERT → Server 03**
- **QuickBooks (combo) → Server 4A**
- **QuickBooks only (heavy) → 5 large servers**
- **Lassert → Multiple servers (shared folder setup, 100-user license)**
- **File servers → FS01 (My CFO), FS02 (My Tax Filer)**
- **Windows 11 systems → Drake + Lassert apps**